

SD-LoRA: Scalable Decoupled Low-Rank Adaptation for Class Incremental Learning

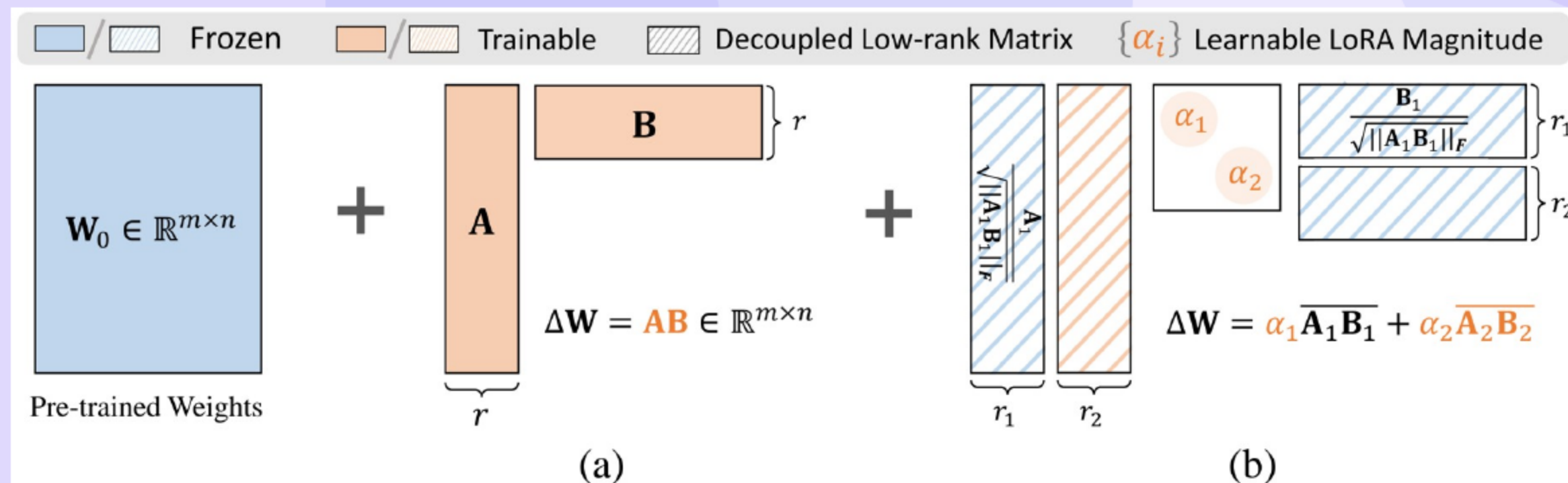
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01 SD-LoRA: Authors and Affiliations

- Author team spans academia and industry across multiple institutions
 - Yichen Wu, Hongming Piao, Long-Kai Huang, Renzhen Wang, Wanhua Li, Hanspeter Pfister, Deyu Meng, Kede Ma, Ying Wei
- Affiliations include CityU HK, Harvard, XJTU, Tencent AI Lab, Zhejiang University, and Pengcheng Laboratory
 - Cross-lab collaboration enables **broad expertise**
- Equal contribution and senior authorship are indicated for clarity
 - Clear roles improve **attribution clarity**

02 SD-LoRA: Scalable Continual Learning without Rehearsal



- Scalable Decoupled LoRA enables class-incremental learning without rehearsal
 - Method decouples magnitude and direction across tasks for **stable reuse**
- Single final model supports task-free inference with no routing
 - Task-ID-free inference; no routing
- Extensive experiments show strong performance across benchmarks and backbones
 - Rehearsal-free across settings
- Acronyms: LoRA=Low-Rank Adaptation; PEFT=Parameter-Efficient Fine-Tuning; ViT=Vision Transformer; KD=Knowledge Distillation; RR=Rank Reduction; AAA=average accuracy across seen tasks

03 Introduction: CL challenges and foundation-model limits

- Continual learning suffers from catastrophic forgetting on sequential tasks
 - Forgetting reduces **retained accuracy**
- Prompt-based methods require retrieval at inference and rehearsal methods require storage
 - Both reduce **scalability and efficiency**
- Target properties include rehearsal-free learning, efficient inference, and end-to-end optimization
 - Design aims at **practical deployment**

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Our proposal and contributions: SD-LoRA

- SD-LoRA decouples magnitude and direction to balance stability and plasticity
 - Reweights past directions while learning a new one for each task
- Inference uses a single final model without task-specific selection
 - This simplifies [serving and maintenance](#)
- Two parameter-efficient variants reduce overhead while preserving accuracy
 - Rank Reduction and Knowledge Distillation offer **compact updates**

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Positioning among CL and PEFT methods (condensed related work)

- Class-incremental learning is addressed without task identifiers
 - Focus avoids **test-time routing**
- Prompt retrieval and rehearsal buffers create scalability and storage trade-offs
 - Trade-offs hinder **deployment at scale**

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SD-LoRA for CL: problem setup and overview

- Tasks arrive sequentially with no access to past data and class-incremental evaluation
 - Setup reflects **realistic constraints**
- LoRA updates modify a low-rank residual while base weights remain frozen
 - Parameter efficiency drives [lightweight training](#)
- SD-LoRA contrasts with vanilla LoRA by reusing fixed directions and new magnitudes
 - Approach focuses on **direction reuse**

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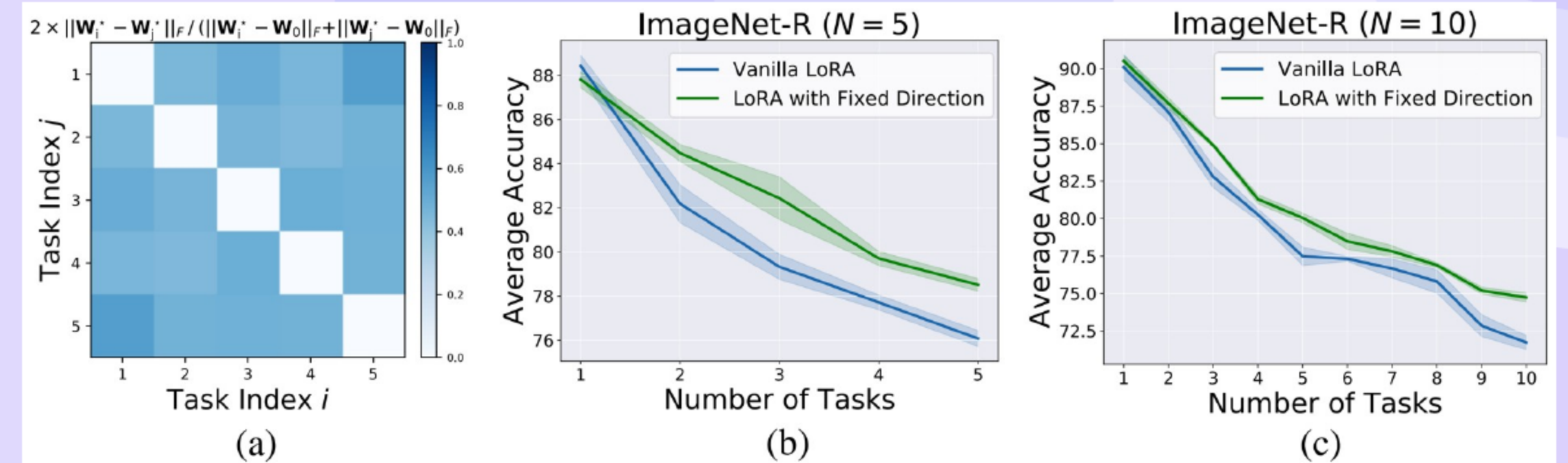
Core mechanism: decouple magnitude and fix directions

- Past directions remain fixed while new magnitudes and one new direction are learned
 - Design preserves **stability across tasks**
- Layer output combines the base weights, scaled past directions, and the current direction
 - Composition achieves [controlled adaptation](#)
- Direction quality is prioritized while magnitudes modulate reuse across tasks
 - Separation improves **learning dynamics**

08 Optimization objectives and training signals

- Stability is achieved by reweighting fixed directions associated with prior tasks
 - Mechanism mitigates **catastrophic forgetting**
- Plasticity is provided by adding one new low-rank direction per task
 - Capacity supports **learning new classes**
- Outcome enables rehearsal-free training and task-agnostic inference
 - Final model requires **no selection**

09 Empirical analysis: direction reuse and low-loss paths



- Task-specific optima are mutually close relative to the foundation model weights
 - Closeness supports **direction reuse**
- Early directions dominate and later magnitudes decline over time
 - Dynamics explain **efficient consolidation**
- SD-LoRA follows overlapping low-loss paths while vanilla LoRA forgets
 - Trajectory reduces **performance drop**

10 Parameter-efficient variants: SD-LoRA-RR and SD-LoRA-KD

- Rank Reduction (RR): progressively lower rank to cut overhead with minor accuracy trade-off.
- Knowledge Distillation (KD): absorb redundant directions to cap parameter growth.

11 Theoretical intuition: why early directions dominate

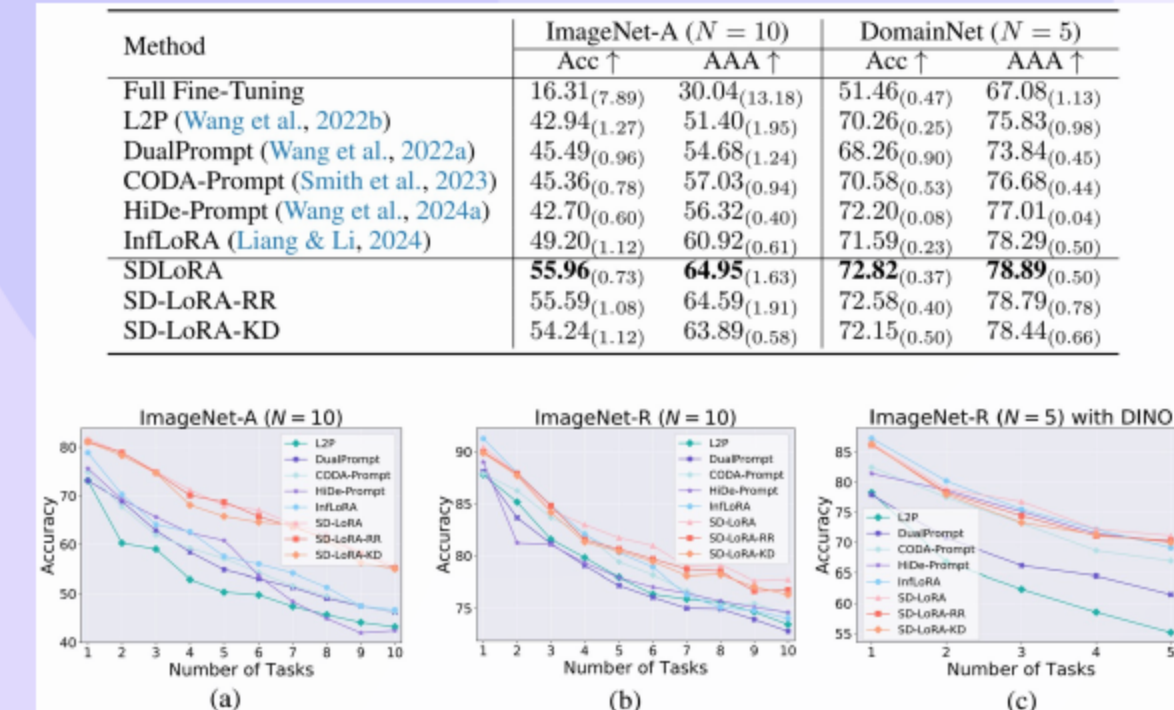
- Gradient descent progressively captures top components of the shared update
 - Process prioritizes **high-utility** directions
- Early directions carry most utility and later ones act as refinements
 - Observation motivates **rank reduction**
- Parameter-efficient designs follow from dominance of early directions
 - Insight supports **knowledge distillation**

12 Experiments: benchmarks, protocol, and setup

Method	ImageNet-R ($N = 5$)		ImageNet-R ($N = 10$)		ImageNet-R ($N = 20$)	
	Acc $\uparrow$	AAA $\uparrow$	Acc $\uparrow$	AAA $\uparrow$	Acc $\uparrow$	AAA $\uparrow$
Full Fine-Tuning	64.92 _(0.87)	75.57 _(0.50)	60.57 _(1.06)	72.31 _(1.09)	49.95 _(1.31)	65.32 _(0.84)
L2P	73.04 _(0.71)	76.94 _(0.41)	71.26 _(0.44)	76.13 _(0.46)	68.97 _(0.51)	74.16 _(0.32)
DualPrompt	69.99 _(0.57)	72.24 _(0.41)	68.22 _(0.20)	73.81 _(0.39)	65.23 _(0.45)	71.30 _(0.16)
CODA-Prompt	76.63 _(0.27)	80.30 _(0.28)	74.05 _(0.41)	78.14 _(0.39)	69.38 _(0.33)	73.95 _(0.63)
HiDe-Prompt	74.77 _(0.25)	78.15 _(0.24)	74.65 _(0.14)	78.46 _(0.18)	73.59 _(0.19)	77.93 _(0.19)
InfLoRA	76.95 _(0.23)	81.81 _(0.14)	74.75 _(0.64)	80.67 _(0.55)	69.89 _(0.56)	76.68 _(0.57)
SD-LoRA	79.15 _(0.20)	83.01 _(0.42)	77.34 _(0.35)	82.04 _(0.24)	75.26 _(0.37)	80.22 _(0.72)
SD-LoRA-RR	79.01 _(0.26)	82.50 _(0.38)	77.18 _(0.39)	81.74 _(0.24)	74.05 _(0.51)	80.65 _(0.35)
SD-LoRA-KD	78.85 _(0.29)	82.47 _(0.58)	77.03 _(0.67)	81.52 _(0.26)	74.12 _(0.66)	80.11 _(0.75)

- Benchmarks include ImageNet-R, ImageNet-A, and DomainNet with varied task splits
 - Settings test **scalability and robustness**
- Metrics report final accuracy and average accuracy across seen tasks (AAA)
 - Evaluation reflects **retention and learning**
- Backbones primarily use ViT-B variants for fair comparison
 - Consistent setup improves **comparability**
- Baselines: L2P (Wang et al., 2022a); DualPrompt (Wang et al., 2022b); CODA-Prompt (Smith et al., 2023); HiDe-Prompt (Wang et al., 2024a); InfLoRA (Liang & Li, 2024); Adapters (Houlsby et al., 2019); Prefix-tuning (Li & Liang, 2021); Full fine-tuning.

13 Results across benchmarks: accuracy and robustness

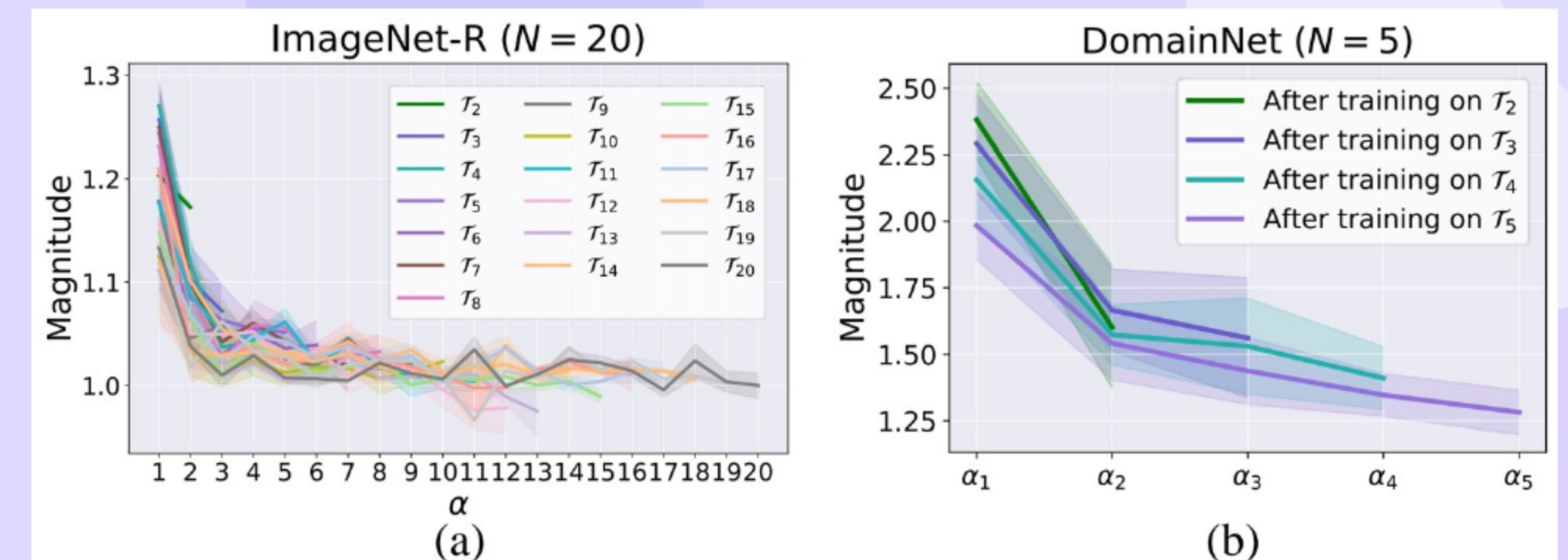


- SD-LoRA outperforms strong baselines across datasets and backbones
 - Gains are **strong across settings**
- Performance margins increase as the number of tasks grows
 - Scalability benefits become **more pronounced**
- Parameter-efficient variants retain accuracy with minimal drop
 - Trade-off remains **favorable**
- Baselines: L2P (Wang et al., 2022a); DualPrompt (Wang et al., 2022b); CODA-Prompt (Smith et al., 2023); HiDe-Prompt (Wang et al., 2024a); InfLoRA (Liang & Li, 2024); Adapters (Houlsby et al., 2019); Prefix-tuning (Li & Liang, 2021); Full fine-tuning.

14 Efficiency and resource footprint

- Computation is comparable to rehearsal-based methods without storing features
 - Eliminating buffers is **memory friendly**
- Parameter count is reduced further with rank reduction while maintaining accuracy
 - Model remains **compact and fast**
- Overall footprint supports deployment under constrained resources
 - Design targets **efficient inference**
- Baselines: L2P (Wang et al., 2022a); DualPrompt (Wang et al., 2022b); CODA-Prompt (Smith et al., 2023); HiDe-Prompt (Wang et al., 2024a); InfLoRA (Liang & Li, 2024); Adapters (Houlsby et al., 2019); Prefix-tuning (Li & Liang, 2021); Full fine-tuning.

15 Limitations and failure modes



- Current evidence focuses on Vision Transformers (ViTs) and specific placements with uncertain transfer to other backbones
 - Limited scope risks **reduced generality**
- Suboptimal early directions can entrench bias and hinder later correction
 - Bias persistence can **degrade performance**
- When task IDs are unavailable, routing-dependent prompt methods can hurt scalability at inference time
- No task-ID or routing required by SD-LoRA, but effectiveness still depends on shared structure across tasks

- Theory should characterize shared low-loss regions and non-linear heads
 - Foundations would **guide design**
- Applicability to adapters, prompts, CNNs, large language models, and multimodal encoders remains open
 - Extensions can **broaden impact**
- Dynamic rank schedules and stronger distillation could improve efficiency and stability
 - Data-driven control offers **adaptive capacity**
- Streaming scenarios with concept drift require task-agnostic adaptation
 - Robustness reduces **drift-induced decay**

Thank You

Questions & Discussion